Đồ án 1

THIẾT KẾ PHẦN CỨNG TĂNG TỐC AUTOENCODER KHỬ NHIỄU ẢNH TRÊN FPGA

Denoiser

https://blog.keras.io/building-autoencoders-in-keras.html

Quá trình train tham khảo từ blog trên cho model Denoiser:

```
Model: "denoiser"
```

Layer (type)	Output Shape	Param #
input (InputLayer)	(None, 28, 28, 1)	0
conv1 (Conv2D)	(None, 28, 28, 4)	40
pool (MaxPooling2D)	(None, 14, 14, 4)	0
conv2 (Conv2D)	(None, 14, 14, 8)	296
upsample (UpSampling2D)	(None, 28, 28, 8)	0
conv3 (Conv2D)	(None, 28, 28, 4)	292
conv4 (Conv2D)	(None, 28, 28, 1)	37

```
 Total params: 665 (2.60 KB)
 Trainable params: 665 (2.60 KB)
 Non-trainable params: 0 (0.00 B)
Total params: 665
```

Với 665 trọng số phù hợp cho thiết kế FPGA.

```
E0000 00:00:1775996483.419146   17040 util.cc:131] oneDNN supports DT_INT32 only on platforms with AVX-512. Falling back to the default Eigen-ba
sed implementation if present.
469/469 ━━━━━━━━━━━━━━━━━━━━ 5s 8ms/step - loss: 0.2179 - psnr_metric: 14.8922 - val_loss: 0.1205 - val_psnr_metric: 17.6890 - learning_rate: 0.
0010
Epoch 2/100
469/469 ━━━━━━━━━━━━━━━━━━━━ 4s 8ms/step - loss: 0.1171 - psnr_metric: 17.9220 - val_loss: 0.1134 - val_psnr_metric: 18.1952 - learning_rate: 0.
0010
Epoch 3/100
469/469 ━━━━━━━━━━━━━━━━━━━━ 4s 9ms/step - loss: 0.1131 - psnr_metric: 18.2308 - val_loss: 0.1111 - val_psnr_metric: 18.3910 - learning_rate: 0.
0010
Epoch 4/100
469/469 ━━━━━━━━━━━━━━━━━━━━ 4s 9ms/step - loss: 0.1114 - psnr_metric: 18.3803 - val_loss: 0.1098 - val_psnr_metric: 18.5044 - learning_rate: 0.
0010
Epoch 5/100
469/469 ━━━━━━━━━━━━━━━━━━━━ 5s 11ms/step - loss: 0.1104 - psnr_metric: 18.4662 - val_loss: 0.1092 - val_psnr_metric: 18.5697 - learning_rate: 0
.0010
Epoch 6/100
469/469 ━━━━━━━━━━━━━━━━━━━━ 9s 8ms/step - loss: 0.1098 - psnr_metric: 18.5205 - val_loss: 0.1089 - val_psnr_metric: 18.5904 - learning_rate: 0.
```

```
n_denoiser_keras.py
469/469 ━━━━━━━━━━━━━━━━━━━━ 4s 9ms/step - loss: 0.1036 - psnr_metric: 19.1335 - val_loss: 0.1029 - val_psnr_metric: 19.1759 - learn
5625e-05
Epoch 98/100
469/469 ━━━━━━━━━━━━━━━━━━━━ 5s 11ms/step - loss: 0.1036 - psnr_metric: 19.1334 - val_loss: 0.1029 - val_psnr_metric: 19.1751 - lear
.5625e-05
Epoch 99/100
469/469 ━━━━━━━━━━━━━━━━━━━━ 5s 11ms/step - loss: 0.1036 - psnr_metric: 19.1336 - val_loss: 0.1029 - val_psnr_metric: 19.1755 - lear
.5625e-05
Epoch 100/100
469/469 ━━━━━━━━━━━━━━━━━━━━ 6s 12ms/step - loss: 0.1036 - psnr_metric: 19.1336 - val_loss: 0.1029 - val_psnr_metric: 19.1751 - lear
.5625e-05

Test MSE: 0.012925
Test PSNR: 18.89 dB
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is c
gacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my
')`.
```

- ⇨ PSNR tăng => Mô hình train hiệu quả.

Kết quả đạt được:
Test MSE: 0.012925

Test PSNR: 18.89 dB

Name	Date modified	Type	Size
denoiser.h5	12/04/2026 7:30 pm	H5 File	43 KB

- ⇨ Sinh ra .h5 để đưa vào Py2C

Classifer – Lenet 5

Model: "denoiser"

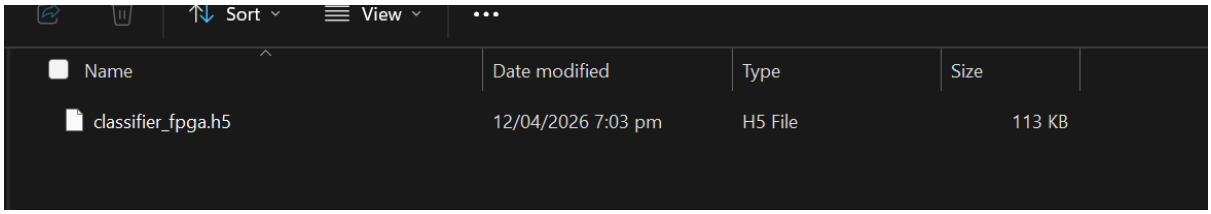
Layer (type)	Output Shape	Param #
input (InputLayer)	(None, 28, 28, 1)	0
conv1 (Conv2D)	(None, 28, 28, 8)	80
pool (MaxPooling2D)	(None, 14, 14, 8)	0
conv2 (Conv2D)	(None, 14, 14, 16)	1,168
upsample (UpSampling2D)	(None, 28, 28, 16)	0
conv3 (Conv2D)	(None, 28, 28, 8)	1,160
conv4 (Conv2D)	(None, 28, 28, 1)	73

```
 Total params: 2,481 (9.69 KB)
 Trainable params: 2,481 (9.69 KB)
 Non-trainable params: 0 (0.00 B)
Total params: 2,481
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow
```

```
Acc (noisy input) : 95.13%
Acc (clean input) : 97.31%
WARNING:absl:You are saving your
acy. We recommend using instead
```

Kết quả train chênh lệch giữa ảnh nhiễu và ảnh sạch là 2.18%

Con số này dùng để chứng minh denoiser có giá trị:

- Không có denoiser: 95.13% (classifier nhận ảnh nhiễu trực tiếp)
- Có denoiser: ~97%+ (classifier nhận ảnh đã khử nhiễu

⇨ Khi denoiser hoạt động tốt → accuracy phải tăng từ 95.13% lên gần 97.31%.